BIOCHEMISTRY

(Group-B)

Clinical Endocrinology

Introduction

1. Define and classify hormone. Describe briefly the mechanism of hormonal action. (Ja-05, 12, Ju-09)
2. Enumerate the different types of cellular communication? Write the brief note on cytokines. (Ju-05)
3. Classify hormone according to chemical nature. (Ja-10, 13)
4. What are the hormones of pituitary gland. (Ja-05, Nov-18)
5. **S.N.** i) Wilson's disease (Ja-17) ii) ACTH (Nov-18)

Growth Hormone

1. Discuss about the feedback control of GH secretion. (Ju-05, 08, Ja-09)
2. Function of GH./metabolic function **(CMU:**May-22) (Ju-11, Ja-06, 07, 08, 09) Difference between somatostatin and somatomedin. (Ja-09)
3. Acromegaly **(CMU**-May-22, 19) (Ju-05, 08, Ja-09, 11, 14, May-18, Nov-18). Mode of action of GH and ADH. (Ju-07, 08, 09)
4. Write down the confirmatory bio-chemical test for GH deficiency, (Ja-09, 08,13, Ju-15) Acromegaly & DM (Ju-13, Ju-15)
5. Difference between Gigantism and acromegaly. (Ja-12)
6. **S.N:** a) Dwarfism. (Ju-07,Ja-06)
   b) Acromegaly(Ja-09, Ju-16, May-19)
   c) Milk ejection reflex.(Ju-08)
   d) Tetany (Nov-17) e) IGF (Nov-18)
   f) Obesity **(CMU**-Nov-20)

Cortisol & Aldosterone

1. Write down the function of aldosterone. **(CMU**-May-21) (Ju-05)
2. Mode of action of aldosterone, calcitonin. (Ja-08, 11)
3. Write down the function of cortisol. (Ja-05,08,10,15)
4. State briefly with a diagram of the synthesis of cortisol with its permissive action. (Ja-11, 12, 17 Ju-08, 07, May-15) What is permissive action.(Ja-15, May-15)
5. **S/N-** Deficiency features of cortisol. (Ju-06)
6. Write down the confirmatory bio-chemical test for Addison's diseases. (Ju-07, Ja-09, 08)
7. How will you differentiate primary and secondary adrenal insufficiency biochemically. (Ju-09)
8. **Mention the principle & procedures of the biochemical test to diagnose the**
   a) Addisson's disease. **(CMU:** May-24) (Ja-07,08, Nov-16)
   b) Acromegaly (Ja-11, May-18, Nov-18)
9. **Mention the biochemical tests for the diagnosis of the-**
   a) Cushing's syndrome **(CMU:** Nov-24, 22) (May-19)
   b) Pituitary dwarifsm. (Ja-12)
   c) Aldosterone escape.(Ja-07)
10. Show with diagram the mechanism of action of cortisol (Ju-13)
11. **SN:** a) Dexamethasone suppression test. (Ju-09, 10, 11, Ja- 14, 17)
    b) Calcitriol (Nov-17)
12. What is aldosteron escape? **(CMU:** May-21)

Thyroid Hormone

1. Name the thyroid hormones (Ja-14, May-15, Ja-16). with normal ranges **(CMU:** May-22) Write the steps the thyroid hormone synthesis with diagram. (Ja-05,06,09,08, Ju-14, Ju-15, Nov-16, May-19)

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2. Function of thyroid hormone. (Ju-06, Ja-08, 09)
3. List the thyroid function test with interpretation **(CMU**-May-21, 20, 19) (Ja-14,15, May-15, Nov-17) Give the clinical importance of each test. (ja-05, 06, 07, 17 Ju-08, 09, 10, 11)
4. Give the reference value of thyroid hormone & TSH level in plasma is primary hypothyroidism and hyporthyroidism (Nov-18). Biochemical pictures of hypothyrodism **(CMU:**May-22)
5. How will you diagnose a case of primary hypothyroidism? (Ja-15) or Laboratory finding of primary hypothyroidism. **(CMU:** May-24)
6. Mention clinical importance of TSH. **(CMU:** May-21, 20)
7. **S/N-** a) Cretinism (Ja-05, 09, Ju-13, May-15, 17, Nov-18)
   b) Myxoedema (Ju-05)
   c) Clinical importance of TSH, C-peptide, IGF-1 (Ja-10)
   d) TSH **(CMU:** Nov-20) (May-16)

ADH

1. Mode of action of ADH. (Ju-08, Ju-10)Name the posterior pituitary hormone. Explain mechanism of action of any one of them with diagram (Ja-16)

Parathyroid Hormone

1. Name the hormone and vitamins that maintain blood Ca^{2+} level. (Ja-05, 08)
2. Write down the mode of action of calcitonin. (Ja-08)
3. **Short note:** a) Tetany (Ja-16, Nov-16) b) Calcium regulating hormons. **(CMU:** May-21)
4. Functions & mode of actions of Ca^{++} regulating hormone. (Ju-09)
5. Discuss with diagram the mechanism of regulation of blood Ca^{++} level. (Ja-10, Nov-18)

Pancreatic Hormone

1. Name the hormone of pancreatic gland. (Ja-07)
2. Mechanism of action of insulin with diagram. (Ju-006, Ja-05, 06, 07, 08, 09, 10, 11, Nov-18)
3. Function of insulin. (Ja-06, 07, 08, 09, Ju-05)
4. Give the pathophysiology of diabetic keto-acidosis. (Ju-05)
5. How OGTT is done? (May-15,17, Ju-15) Preparation, interpretation, indication of OGTT **(CMU**-Nov-22, (May-15, 16, Nov-16) Procedure (May-18). What is OGTT? **(CMU**-Nov-20,May-20)
6. Show a diagram the metabolic consequences due to insulin deficiency in diabetes mellitus. (Ja- 17 May-15, Nov.-16) What is alimentary glycosuria **(CMU:** Nov-22)
7. Normal values and physiological importance of fasting and post prandial blood glucose. (Ju-09)
8. Write a plan to diagnose a case of DM. (Ju-07, May-17)
9. How normal blood glucose level is maintained. Give the diagnostic criteria of diabetes mellitus. (Ja-12, 14) Write normal blood glucose level (Ja-14), What is gastational DM (May-15)
10. What are the metabolic goal of management of diabetes mellitus? Mention the clinical importance of IGT and IFG (Impaired fasting glucose). (Ju-11)
11. Write the WHO criteria for diagnosis DM. **(CMU:** Nov-21, 20, May-21)
12. Write down the mode of action of insulin **(CMU:** May-21) (Ju-15). What are the changes occur in diabetic ketoacidosis. (Ju-14, Ju-15)
13. Give the blood picture of diabetic ketoacidosis. What is hypoglycemia.(Ju-16)
14. What are indication and various interpretation of OGTT. (July-16)
15. Give the arterial blood picture values of diabetic ketoacidosis. (May-16)
16. Diabetic ketoacidosis. **(CMU**-Nov-20)
17. How will you diagnose diabetes mellitus biochemically in pregnant & non pregnant subjects? **(CMU**-May-20)
18. **S.N :-** a) Glycaemic index. (Ja-17) b) Insulin resistance (Nov-15)
    c) OGTT **(CMU**-May-19) (Nov-18) c) GMD **(CMU**-Nov-21)
19. What is insulin resistance? **CMU:** May-21)
20. Discuss the role of insuline maintaining blood glucose level. **(CMU:** May-24)

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Clinical Biochemistry

1. **Give the confirmatory biochemical test of-**
   a) Dwarfism/GH deficiency test. (Ja-08,07)
   b) Dexamethasone test. (Ju-09, 10, 11,Ja-13)
   c) Addison's disease. (Ja-11,15)
   What types of electrolyte & pH imbalance occure in this case? **(CMU:** May-23)
   d) Short & long ACTH stimulation test. (Ju-10,Ja-13)
   e) Thyroid function test with clinical importance.(Ja-13)
   f) Acromegaly. (Ja-11, Ju-14)
   g) Pituitary dwarf. (Ja-12)
   h) Give the normal value of lipid profile & their importance **(CMU:**Nov-19) (Nov-18)
2. Outline to diagnose a case of DM? (Ja-07,12) What is glucostat **(CMU:** Nov-22)
3. Give the chemical abnormalities of diabetic ketoacidosis in blood? How OGTT is done? (Ja09, 07)
4. Give the diagnostic criteria of DM in pregnant & non pregnant woman? (Ja-08, 09, 11,13, Ja-16)
5. Mention the cousequence due to insulin deficiency in DM with diagram?
6. List the liver function test with values. **(CMU:**Nov-22, May-21, 19), (Ju-14, 16, Nov-17).
   With their interpretation **(CMU-**May-20, 19), (May-17) Under three broad headlines "Hepatocellular, Synthetic & Excretory" Function tests. **(CMU:** May-23)
7. Classify hyperlipidemia with normal value of lipid profile. (Ju-06, 10)
8. Give the values of plasma protein. List their important function? (Ja-16)
9. What is jaundice? Classify the jaundice with clinical interpretations **(CMU-**May-19) (Nov-17, 19)
10. How will you investigate a case of jaundice? (Ja-08, Nov-15, Ja-16).
11. Difference between hemolytic **(CMU-**May-19) and latent jaundice. (Nov-15)
12. List complication of DM **(CMU-**May-19)
13. List the necessary investigation for a case of jaundice (May-18)
14. Common cause of hyperlipidemia **(CMU-**May-19) (Nov-15)
15. **Give the normal value of following-**
    a) FBG, IFG, IGT, post prandial blood glucose. (Ju-09, 10)
    b) BUN
    c) Blood urea level
    d) Serum creatinine (Ju-10)
    e) Serum bilirubin (Nov-18)
    f) ALP/AST
    g) Serum Ca^{++} (Ju-10,Ja-13)
    h) Serum K^{+}(Ju-10,Ja-13)
    i) Lipid profile.(Ja-13, Nov-15)
    j) Blood glucose.(Ja-13)
    k) OGTT **(CMU-**May-19) (Ju-14, Nov-17, 18)
    l) GDM (Ja-16)
    m) Latent Jaundice (Nov-15)
    n) Lipid lowering drugs (Nov-15)
    o) Prothrombin time (Nov-18)
    p) Anion gap **(CMU-**Nov-19), (Nov-18, May-19)
16. Clinically important enzyme with their tissue location. (Nov-15)
17. **Mention the biochemical pictures of-**
    a) Diabetic ketoacidosis (May-18) b) Cushing's syndrome
18. State the biochemical change of blood & urine in Diabetic ketoacidosis (Ju-13)
19. Name the RFT with their normal value & mention their clinical importance (Ju-13)
20. Biochemically how will you diagnose a case of hepatocellular jaundice. (Ju-14)

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21. How plasma enzyme help in diagnosis of Myocardial infraction (Nov-15)
22. Name the clinical importance of CPK, MB and LDH. (Ju-16)
23. State the importance of LDH. (Ju-16)
24. **SN:** Proteinuria **(CMU**-Nov-21, May-20, 19) (Nov-18),
    a) Common clinical chemistry tests done in clinical laboratory. (May-16)
    b) Serum sodium ion (Ju-16) * Photometry
    c) Name cardiac enzyme and marker used to diagnose MI. Mention their appearance. **(CMU**-Nov-19)
    d) Pyruvate dehydrogenase complex. (Ju-16)
    e) Enzymes of clinical significance. **(CMU**-May-20)
    f) Cushing syndrome (Ja-12, Ju-14, Ja-16) **(CMU**-May-20)
    g) Benedict's test **(CMU**-Nov-21)
    h) Cardiac marker **(CMU**-Nov-21)
25. List the biochemical markers of acute myocardial infarction with their normal values. (Nov.16)
26. Define jaundice. State the difference among different types of jaundice. **(CMU:**Nov-21)
27. What is edema **(CMU**-May-21)
28. What is the fredrickson type II hyperlipidemia? **(CMU:** Nov-23)
29. By which tests sugar & protein are detected in a sample? **(CMU: May-25)** Write down the underlying principle & significance of their presence in urin. **(CMU**: May-23)
30. What is normal urinary Albumin Exercation Rate (AER)? Mention the types of albuminusia with AER. Name two external diseases causing albuminuria. **(CMU:** May-24)
31. What are the major causes of elevated levels of blood AST and ALT? **(Nov-25)**

Genetics & Molecular Biology

1. Give the steps of protein synthesis with diagram? **(May-25) (CMU:** Nov-21)(Ju-05, 07, 08, 11, Ja-10, 13, Nov-18) Name the purine & pyrimidine base **(CMU:** Nov-21) (Ja-14)
2. **Give the criteria of-**
   a) Autosomal dominant disorder with example and criteria. **(CMU:** Nov-23, 22, May-20) (Ju-10,11,14, 15, 16 Ja-14, May-15,17 , 18, Nov-16, 18)
   b) Autosomal recessive disorder. **(CMU:** May-23, 22, Nov-21) (Ja- 11, 12, 13, 17, Nov-17)
3. Difference between DNA & RNA. **(CMU:** Nov-21) (Ja-14,15, Ja-16)
   Function of DNA & RNA. (Ja-15)
4. **SN –** a) PCR **(CMU**-May-19) (Ja-09,08,11,12, May-19) with technique **(CMU: Nov-23)**
   b) Recombinant DNA technology. (Ju-10,14,15 Ja-13,15, May-16, 17)
   c) Allele. **(CMU:**Nov-22, 19) (Ju-09,11,Ja-07,13,15, Nov-16)
   d) Nonsense mutation. (Ja-07)
   e) Non-dysjunction **(CMU**-May-22) (May-16)
   f) Intron (Ja-06)
   g) Mosaicism (Ju-14, May-15, Nov-17)
   h) Down's syndrome (May-16, Nov-17)
   i) Sex linked disorder (Ja-06)
   j) Trineucleotide repeats
   z) Go stage.
   k) Deletion (Ja-09)
   l) Karyotyping **(CMU:** May-22) (Ju-10,11,14, Ja-15, Ju-15)
   m) Genetic counselling (Ja-11,12)
   n) Codon (Ja-11, May-18)
   o) RFLP **(CMU:** May-22,19, Nov-19) (Ju-11,13,14, Ja-16)
   p) Mutation (Ja-17, Ju-08, Nov-15)
   q) Trisomy 21 **(CMU**-Nov-19) (Ju-13, 16)
   r) Chromosomal Abnormality. **(CMU**-May-21, Nov-19) (Ja-15, Ju-15, May-19)
   s) Mendel's Law (Nov-16) inheritance of madanls law **(CMU:**Nov-22)
   t) Law of segregation (May-16)
   u) 45XO **(CMU**-May-20) (Ju-16)
   v) Karyotype **(CMU**-Nov-19, May-19) (Nov-16, Ja-17) of down syndrome, klinefelter's

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& turner's syndrome (**CMU:** Nov-22)

w) Translation
x) Turner syndrome (May-18)
y) Cell cycle (**CMU:** Nov-22, 20)
z) 47XXY (**CMU**-May-20)

5. Sex linked disorder. (Ju-06) Criteria of X-linked recessive disorder with example.(**Nov-23,** Ja-12)
6. Define & classify mutation. Discuss various types of mutation / point mutation with example. **(Nov-25)** (**CMU**-May-20, 19) (Ja-12, Nov-15, 17, May-16, Ju-16)
7. State the steps of polymerase chain reaction **(Nov-25)** & give its importance (Ju-13, Ja-15, Nov-16)
8. Define genetic code & write characteristics of genetic code/ Explain the properties of genetic code. (**CMU: May-25**, 20, Nov-19) (Ja-14,15, Nov-15, May-18)
9. What is recombinant DNA technology? Mention its importance. (**CMU: May-25,** 23, Nov-21, 20) (Ju-14, Ja-17)
10. What do you mean by autosomal dominant disorder? (Ju-14, Nov-16)
11. What do you mean by central dogma of life (**Nov-25,** 15) What is allele (**CMU:** May-22)
12. Criteria of DNA replication (Nov-15). How it occurs in cell (Nov-15)
13. Difference of biological cloning and PCR (**CMU:** May-23, Nov-20) (Nov-15)
14. Define polymorphism (**CMU**-Nov-22) (Nov-15)
15. What do you mean by RFLP and VNTR (Nov-15) Write down their use. (Nov-15)
16. What is biological cloning? give it's basic step. (Nov-15)
17. Explain post translational modification with diagram. (**CMU**-May-19) (Ja-16) Give three example **(CMU: May-25)**
18. List the criteria of X-linked recessive disorders with example (May-18). Give example of autosomal recessive disorder (May-16)
19. Define DNA Replication & Give the steps of DNA replication? (**CMU**: May-22) (May-17) With enzyme in this procedure. **(CMU:** Nov-23)
20. What is covalent modification? **(CMU:** May-19) (May-17)
21. Give the blood picture of diabetic ketoacidosis what is hypoglycemia. (Nov-17)
22. State the law of mendel. Explain dominant state of allele with example (May-18)
23. Give the steps of transcription (**CMU**-Nov-22) (May-18)
24. Name the nitrogenous base. What are their function (Nov-18)
25. Define genetic code and codes. Write down the criteria of genetic code. (**CMU:**Nov-21)
26. List the requirement for translation.
27. Define codes (**CMU:**May-21)
28. Define chromosome (**CMU:**May-21) Classify chromosomal disorder **(CMU: May-23)**
29. Classify point mutation with example (**CMU**-Nov-21)
30. Write down the importance of PCR technique **(CMU:** Nov-22)
31. What is hybrid DNA? **(CMU:** May-23)
32. Classify genetic disorder. **(CMU:** Nov-23)
33. What happens in nonsense mutation. Give example. **(CMU: May-25)**
34. Write down the characteristics of plasmid as vector? **(May-25)**
35. What disease are diagnosed y PCR? **(Nov-25)**

নিশ্চয়ই আল্লাহ ঐসব লোকদের পছন্দ করেন, যারা তাঁরা পথে এমনভাবে কাতারবন্দী হয়ে লড়াই করে, যেন তারা সীসা ঢালা প্রাচীর। -আল কুরআন (৬১:০৪)

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